

Scientific and Regulatory Evidence Base for Sweeteners in the Israeli Retail Market

A. Shelf Reality and Retail Availability

Authorisation and Physical Presence of Novel Sweeteners

Establishing the baseline of what is legally permitted and physically accessible on the shelves of Israeli retailers is a prerequisite for any analytical evaluation of the sweetener market.

D-allulose (also referred to as D-psicose) is legally authorised for use in the State of Israel¹. On February 1, 2024, the Israeli Ministry of Health (MoH) National Food Services published a formal directive approving D-allulose as a "Novel Food" (מזון חדש). This regulatory approval is not open-ended; it strictly limits the incorporation of D-allulose to specific food categories and mandates precise maximum concentrations¹:

- **Yogurts and flavored yogurt drinks:** Up to $5 \text{ g}/100 \text{ g}$ of finished product¹.
- **Cream cheeses, flavored cheeses, mousse, and whipped cream powder:** Up to $5 \text{ g}/100 \text{ g}$ ¹.
- **Ice creams, popsicles, and sorbets:** Up to $5 \text{ g}/100 \text{ g}$ ¹.
- **Puddings, jams, and confitures:** Up to $10 \text{ g}/100 \text{ g}$ ¹.
- **Cakes, cookies, savory pastries, baked snacks, and baking mixes:** Up to $10 \text{ g}/100 \text{ g}$ ¹.
- **Chewing gum:** Up to $50 \text{ g}/100 \text{ g}$ ¹.
- **Candies:** Up to $25 \text{ g}/100 \text{ g}$ ¹.
- **Cereal bars and breakfast cereals:** Up to $2 \text{ g}/100 \text{ g}$ ¹.
- **Carbonated beverages:** Up to $2.1 \text{ g}/100 \text{ mL}$ ¹.
- **Fruit juices, nectars, coffee drinks, and milk substitutes (such as soy drinks):** Up to $3.5 \text{ g}/100 \text{ mL}$ ¹.

The MoH strictly prohibits the addition of D-allulose to any food category not explicitly defined in the directive, with a blanket prohibition on its use in infant formula, baby foods, and designated medical foods¹. In terms of retail shelves, D-allulose is physically present within Israeli retail³. It is sold as a raw sweetening agent in specialty health food stores, local nut and

seed retail chains, and is accessed by Israeli consumers via direct-to-consumer online import platforms³.

Monk fruit (derived from *Siraitia grosvenorii* and characterized by its primary sweet component, mogroside V) is legally authorised and physically present in Israel⁶. In 2017, the Israeli MoH issued technical guidelines to food manufacturers permitting the use of "concentrated monk fruit juice" (מיץ מרוכז מפרי המונק) as a food ingredient⁶. As with D-allulose, the MoH applies a precautionary exclusion on its use in infant and baby foods due to a lack of developmental toxicological literature in very young cohorts⁶. Monk fruit is physically present on Israeli retail shelves, primarily sold to consumers as a tabletop sweetener (frequently blended with erythritol to allow 1:1 volume measurements matching sucrose) or as concentrated liquid drops in retail channels⁸.

Sweeteners Listed on Israeli Product Labels

To guide consumer-facing analyses, the specific high-intensity sweeteners, rare sugars, and polyols that legally appear on ingredient lists in Israeli retail must be identified. Israeli food labeling regulations align with European standards, requiring the declaration of either the specific name of the food additive or its corresponding E-number¹³.

Sweetener Category	Chemical Name	Hebrew Name on Label	E-Number
High-Intensity (Synthetic)	Acesulfame Potassium	אצסולפאם ק"י / אצסולפאם K	E950 ¹⁵
	Aspartame	אספרטיים	E951 ¹⁵
	Cyclamic Acid (and salts)	ציקלאמט / חומצה ציקלאמית	E952 ¹⁵
	Saccharin (and salts)	סחרין	E954 ¹⁵
	Sucralose	סוכרלוז	E955 ¹⁵
	Neotame	נאוטם	E961 ¹⁵
	Aspartame-Acesulfame Salt	מלח אספרטיים-אצסולפאם	E962 ¹⁵

	Advantame	אדוונטם	E969 ¹⁵
High-Intensity (Natural)	Steviol Glycosides	גליקוזידים של סטביל	E960 ¹⁵
	Thaumatococin	תאומטין	E957 ¹⁵
	Neohesperidine DC	נאוהספרידין די.סי	E959 ¹⁵
Sugar Alcohols (Polyols)	Sorbitol	סורביטול	E420 ¹⁹
	Maltitol	מלטיטול	E965 ¹⁹
	Xylitol	קסיליטול	E967 ¹⁹
	Erythritol	אריטריטול	E968 ¹⁹
Unnumbered Sweetening Agents	D-Allulose	אלולוז	N/A (Labeled as "אלולוז") ¹
	Monk Fruit Extract	תמצית פרי הנזיר / מיץ מרוכז מפרי המונק	N/A ⁶

B. Per-Sweetener Evidence

1. Steviol Glycosides (E960)

The European Food Safety Authority (EFSA), the Joint FAO/WHO Expert Committee on Food Additives (JECFA), and the Israeli Ministry of Health demonstrate complete regulatory alignment on steviol glycosides²⁰. All three bodies establish an Acceptable Daily Intake (ADI) of **4 mg/kg · bw/day** expressed as steviol equivalents²⁰.

Human interventional evidence from randomized controlled trials (RCTs) indicates that steviol glycosides generally exert a neutral effect on postprandial glucose levels and short-term glucose tolerance²¹. In a key RCT by Suez et al. (2022), the administration of stevia sachets over 14 days at doses below the ADI in healthy adults resulted in neutral glycemic responses compared to vehicle-only controls²².

However, the study revealed significant, longitudinal profiles of oral and fecal microbiome alterations unique to stevia consumers²³. These gut microbial changes are characterized by high responder heterogeneity, meaning that the specific taxa altered vary significantly between individuals based on baseline microbiome configurations²².

Long-term observational studies associate high-intensity sweetener intake with obesity, but interventional models show that steviol glycosides do not causally impair systemic metabolic markers at approved levels²¹.

2. Sucralose (E955)

The regulatory consensus on sucralose exhibits minor historical divergences in hazard evaluation. JECFA and the Israeli MoH authorize an ADI of $15 \text{ mg/kg} \cdot \text{bw/day}$ ²⁵. EFSA similarly approved the safety of sucralose, although its published scientific reviews have evaluated dietary exposure models against a more conservative threshold of $5 \text{ mg/kg} \cdot \text{bw/day}$ ¹⁵.

Human interventional trials have demonstrated that sucralose is not metabolically inert. In the Suez et al. (2022) RCT, healthy adults consuming sucralose for 14 days exhibited significant impairment in postprandial glycemic tolerance compared to baseline and vehicle controls²². Metagenomic and functional analysis of the fecal microbiome showed that sucralose distinctly altered gut microbial composition and significantly modified functional pathways, such as purine metabolism²².

These metabolic and glycemic shifts are highly individual-dependent; when fecal microbiota from human "top responders" (those with the most pronounced glycemic impairment) were transplanted into germ-free mice, the mice developed glucose intolerance, whereas transplants from non-responders or controls did not alter glycemic tolerance²².

This provides causal, interventional evidence of a personalized, microbiome-mediated glycemic response to sucralose²².

3. Aspartame (E951)

Regulatory authorities maintain complete consensus on the safe consumption limits for aspartame, with JECFA, EFSA, and the Israeli MoH affirming an ADI of $40 \text{ mg/kg} \cdot \text{bw/day}$ ¹⁹.

Human interventional evidence from short-term RCTs shows that aspartame does not impair glucose tolerance or affect insulin sensitivity in healthy populations²². Metagenomic sequencing in RCTs indicates that aspartame alters fecal microbiome composition in an individual-specific manner, though these shifts are less pronounced and distinct from the substantial glycemic-altering patterns associated with sucralose or saccharin²².

In July 2023, the International Agency for Research on Cancer (IARC) and JECFA released concurrent statements regarding aspartame's safety profile²⁶.

- **IARC Statement:** The IARC classified aspartame as **Group 2B ("possibly carcinogenic to humans")**²⁶. This classification was based on "limited evidence" for cancer in humans

(specifically hepatocellular carcinoma), "limited evidence" in experimental animals, and "limited evidence" regarding potential oncogenic mechanisms²⁷.

- **JECFA Statement:** JECFA reaffirmed the ADI of $0 \text{ to } 40 \text{ mg/kg} \cdot \text{bw/day}$ ²⁷. The committee concluded that the evidence linking aspartame consumption to cancer in humans was "not convincing" and that the data evaluated did not justify a modification of the established ADI²⁷.

These two statements represent different scientific assessments: IARC performed a *hazard identification* (determining whether aspartame can theoretically cause cancer under any condition or dose), while JECFA performed a *risk assessment* (determining the probability that cancer or other adverse effects will occur under real-world dietary exposure scenarios)²⁶. Together, they mean that while a theoretical hazard has been flagged, standard dietary consumption of aspartame within the approved ADI of $40 \text{ mg/kg} \cdot \text{bw/day}$ does not present a valid safety concern for the general public²⁶.

4. Acesulfame Potassium (E950)

The regulatory thresholds for acesulfame-K show minor divergences: JECFA has established an ADI of $15 \text{ mg/kg} \cdot \text{bw/day}$, whereas EFSA maintains a more conservative ADI of $9 \text{ mg/kg} \cdot \text{bw/day}$ ¹⁸. The Israeli MoH permits the use of acesulfame-K under standard category limits¹⁶.

There is a lack of isolated, high-quality human interventional RCTs evaluating the direct impact of acesulfame-K on glucose tolerance and the gut microbiome in humans³⁰. Most human interventional data evaluate acesulfame-K in combination with aspartame or other sweeteners, preventing isolated causal attribution³⁰.

In observational epidemiological cohorts (e.g., studies evaluating additive intake and body mass index), higher consumption of acesulfame-K is associated with normal-weight-exceeding status and lower adherence to healthy dietary patterns¹⁷. However, these observational associations do not establish a causal relationship due to the potential for reverse causality¹⁷.

5. Saccharin (E954)

EFSA and JECFA demonstrate regulatory consensus, setting the ADI for saccharin at $5 \text{ mg/kg} \cdot \text{bw/day}$ ¹⁵, which is mirrored by the Israeli MoH.

Human interventional evidence shows conflicting outcomes²². In the RCT by Suez et al. (2022), healthy adults consuming saccharin at doses lower than the ADI for 14 days exhibited significantly impaired postprandial glycemic responses²². Metagenomic analysis showed distinct longitudinal alterations in the composition of both the oral and fecal microbiomes, and fecal transplantation of responder microbiomes into germ-free mice successfully replicated the impaired glycemic phenotype, demonstrating a causal, microbiome-mediated effect²². Conversely, a separate double-blind, parallel-arm human RCT involving 46 healthy adults administering high-dose saccharin reported no significant modifications in glucose tolerance,

insulin sensitivity, gut hormone secretion, or gut microbial diversity/composition²².

These conflicting results indicate that the glycemic and microbial impact of saccharin is highly heterogeneous and sensitive to individual baseline microbiota configurations²².

6. Erythritol (E968)

EFSA, JECFA, and the Israeli MoH maintain complete consensus, classifying the ADI of erythritol as "not specified" or "not limited" due to its extremely low systemic toxicity in standard evaluation models¹⁸.

Human interventional RCTs show that erythritol is glycemic-neutral and does not elevate blood glucose or insulin concentrations³³. While clinical interventional studies in humans show no significant modifications in overall gut microbial diversity³³, ex vivo studies using clinically predictive gut simulation models (such as the SIFR technology) suggest that erythritol may act as a prebiotic by significantly increasing the abundance of beneficial, butyrate-producing microbial families (specifically *Eubacteriaceae* and *Barnesiellaceae*)³⁴.

The Erythritol Cardiovascular and Thrombosis Signal

In 2023, Witkowski et al. published an observational and mechanistic study linking circulating erythritol to cardiovascular events³⁵. Fasting plasma erythritol levels in a clinical discovery cohort (n=1,157) were associated with a three-year risk of incident major adverse cardiovascular events (MACE)³⁶.

This association was confirmed in validation cohorts from the US (n=2,149) and Europe (n=833), where individuals in the highest quartile of plasma erythritol had a two-fold higher risk of MACE³⁶.

Mechanistic tests showed that:

- Exposing human platelets to physiological concentrations of erythritol enhanced stimulus-dependent calcium mobilization, dense granule release (serotonin), and aggregation³⁶.
- Increased circulating erythritol accelerated carotid artery thrombosis and vessel occlusion in animal models (mice)³⁶.
- In a human intervention arm (n=8), ingestion of a ^{30 g} erythritol-sweetened beverage caused a >1000-fold increase in plasma erythritol concentrations (peaking at $\sim 6480 \mu\text{M}$ vs. $\sim 3.8 \mu\text{M}$ baseline) which remained elevated for up to two days, exceeding the threshold needed to alter platelet reactivity³⁶.

Independent cohort data (such as Wang et al., 2019) has shown a historical correlation between elevated erythritol levels and incident coronary heart disease risk using serum samples collected between 1987 and 1989³⁹.

However, critical clinical analyses show that these observational studies cannot prove causality⁴¹. Erythritol is endogenously synthesized by human cells via the pentose phosphate pathway (PPP), which is systematically up-regulated in patients experiencing ischemia, metabolic syndrome, and tissue reperfusion injury³⁵.

Therefore, fasting plasma erythritol levels in cardiac patients likely represent a biomarker of

underlying metabolic stress rather than a consequence of dietary intake⁴³.

Furthermore, the **30 g** dose used in the interventional studies represents a high bolus exposure typical of dietary supplements, rather than standard dietary intake⁴⁵. At normal dietary levels, erythritol is well tolerated and is not associated with altered platelet reactivity³³. Neither the FDA nor EFSA has responded with regulatory restrictions, and erythritol remains classified as a safe sweetener⁴¹.

7. Xylitol (E967)

Regulatory authorities demonstrate alignment, declaring the ADI for xylitol as "not specified"¹⁸. Human interventional trials show that xylitol consumption is glycemic-neutral and does not cause postprandial glucose excursions³⁹. Decades of clinical literature support its efficacy in reducing dental caries and plaque accumulation by inhibiting pathogenic oral biofilms³⁹.

Large single doses exceeding **30 g** can cause osmotic gastrointestinal side effects (nausea, bloating, colic, and watery stools) due to incomplete small-intestinal absorption³³.

The Xylitol Cardiovascular and Thrombosis Signal

In 2024, Witkowski et al. published a follow-up study associating circulating xylitol with cardiovascular risks⁴⁷. Fasting plasma xylitol levels in a diagnostic discovery cohort (n=1,157) were linked to a three-year incident MACE risk⁴⁵.

This association was confirmed in a validation cohort, showing that elevated fasting xylitol was associated with a 57% increase in MACE risk⁴⁵.

The mechanistic and interventional studies showed that:

- Xylitol enhanced multiple indices of platelet reactivity and thrombus formation in vitro and in animal models at baseline fasting concentrations⁴⁵.
- In a human pilot study (n=10), consumption of a **30 g** xylitol-sweetened beverage raised plasma concentrations to a peak of $\sim 630 \mu\text{M}$ at 30 minutes, resulting in an acute increase in stimulus-dependent platelet aggregation⁴².

Similar to erythritol, these findings are subject to confounding⁴². Fasting plasma xylitol in high-risk patients reflects endogenous synthesis through the up-regulated PPP under metabolic stress, not dietary intake⁴².

The interventional study administered a massive **30 g** single dose, which is associated with gastrointestinal side effects and exceeds recommended consumer exposure⁴⁵.

Under normal dietary conditions (such as chewing gum containing **< 10 g/day** consumed in divided **1 to 2 g** doses), xylitol has not been shown to cause systemic accumulation or thrombotic risk⁴⁵.

Neither EFSA nor the FDA has issued warning statements or modified the regulatory status of xylitol⁴¹.

8. Sorbitol (E420)

Sorbitol is approved by EFSA, JECFA, and the Israeli MoH with an ADI classified as "not specified"¹⁸.

Human interventional trials demonstrate that sorbitol has a minimal effect on postprandial glucose excursions¹⁹. Because sorbitol is incompletely absorbed in the small intestine, it enters the colon where it acts as an osmotic agent and undergoes fermentation by the microbiota¹³. There is no high-quality human interventional evidence demonstrating positive, target-specific shifts in the human gut microbiome composition¹⁸. At elevated levels of consumption, sorbitol causes dose-dependent gastrointestinal distress, bloating, flatulence, and osmotic laxation¹³.

9. Maltitol (E965)

Maltitol is approved with an ADI classified as "not specified" by EFSA, JECFA, and the Israeli MoH¹⁸.

Unlike erythritol, maltitol is partially absorbed and metabolized, providing approximately **2.0 to 2.4 kcal/g**¹⁹. Human interventional evidence indicates that maltitol triggers a low but distinct postprandial glycemic response, making it less glycemic-neutral than high-intensity sweeteners or other polyols¹⁹.

Unabsorbed fractions undergo colonic fermentation, resulting in significant gastrointestinal side effects (bloating, abdominal cramps, and osmotic diarrhea) at high doses¹³. High-quality human interventional trials evaluating specific prebiotic gut microbiome benefits are insufficient to demonstrate clinical efficacy¹⁸.

10. Monk Fruit Extract (Mogroside V)

The regulatory status of monk fruit extract displays significant regional divergence⁴⁹:

- **EFSA:** Monk fruit extract is **not authorised** as a food additive or novel food in the European Union¹⁸. The EFSA Panel on Food Additives and Flavourings (2019) concluded that the available toxicity database was insufficient to establish the safety of monk fruit extract as an intense sweetener¹⁸.
- **JECFA:** A definitive safety evaluation and numerical ADI have not been finalized¹². Although initially proposed for evaluation, lack of submitted data historically delayed the establishment of a JECFA ADI⁵².
- **US FDA:** Permitted under the GRAS framework via multiple "no questions" letters for specific preparations⁴⁹.
- **Australian and New Zealand Food Standards (FSANZ):** Approved as an intense sweetener with an ADI of "not specified" under GMP conditions⁴⁹.
- **Israeli Ministry of Health:** Authorised as a food ingredient/concentrate since 2017⁶.

Human interventional evidence from small-scale, short-term RCTs indicates that monk fruit extract can reduce postprandial blood glucose excursions by 10% to 18% and insulin responses by 12% to 22% compared to sucrose⁵⁰.

There is currently insufficient high-quality human interventional evidence evaluating the impact

of pure monk fruit extract on the human gut microbiome, with existing literature limited to ex vivo models⁵⁶.

11. D-Allulose

D-allulose displays significant regional regulatory divergence⁴⁹:

- **EFSA:** D-allulose is **not authorised** for use in the EU and is currently undergoing evaluation under the Novel Food Regulation⁵⁷.
- **JECFA:** JECFA has not established an ADI for D-allulose itself, though it has evaluated the safety of specific enzymatic preparations used in its manufacturing (such as D-allulose 3-epimerase), determining that the manufacturing process does not present safety concerns⁵⁹.
- **US FDA:** Authorised via multiple GRAS notifications, permitting its use across multiple categories⁵⁸.
- **FSANZ:** Approved as a novel food with an ADI of "not specified"⁶⁰.
- **Israeli MoH:** Authorised as a Novel Food as of February 2024 with specific category limits¹.

Human interventional and pharmacokinetic evidence shows that approximately 70% to 80% of an oral dose of D-allulose is rapidly absorbed in the small intestine and excreted unchanged in the urine, avoiding systemic metabolism for energy and contributing a negligible caloric value of **0.4 kcal/g**¹.

Human interventional trials show that D-allulose is glycemic-neutral and does not stimulate postprandial insulin secretion³⁴.

Ex vivo models suggest that the unabsorbed fraction reaching the colon has prebiotic potential, significantly increasing the abundance of beneficial microbial families and raising butyrate production³⁴.

However, human interventional tolerance studies show that high acute ingestion levels cause osmotic gastrointestinal side effects⁶⁰. The lowest single dose associated with these mild

gastrointestinal symptoms in humans is **0.4 g/kg · bw** (equivalent to **28 g** for a **70 kg** adult), with severe diarrhea occurring at **0.5 g/kg · bw**⁶⁰.

Repeated daily intakes exceeding **0.9 to 1.0 g/kg · bw** can result in severe nausea, abdominal pain, and regular diarrhea⁶⁰.

C. Consumer Confusion

A primary risk in consumer education is the potential for consumer confusion between sweeteners of distinct chemical origins, such as stevia, sucralose, and erythritol.

A comprehensive search of scientific databases, public health reports, and Israeli consumer metrics yields **no empirical evidence** confirming that Israeli consumers confuse stevia, sucralose, and erythritol. There are no published surveys, documented Hebrew search engine logs, or rising-query data demonstrating systematic retail confusion among these three

sweeteners in Israel.

The lack of empirical evidence indicates that assertions of consumer confusion in the Israeli retail market are speculative and currently unsupported by objective data.

D. Reformulation Dynamics

On January 1, 2022, Israel introduced a purchase tax on sweetened beverages to encourage public health⁶⁶. High-sugar beverages (exceeding 5 g of sugar per 100 mL) were taxed at 1.0 NIS per liter, while sugar-reduced and diet beverages containing alternative sweeteners were taxed at 0.7 NIS per liter⁶⁸. This policy resulted in a retail price increase of 3.03% to 38.89% and a corresponding decrease in the sales of high-sugar and diet beverages⁶⁷.

In March 2023, the Minister of Finance officially repealed the sweet beverage tax⁶⁶. Following the repeal, retail prices of these beverages did not drop back to their pre-tax levels, remaining 6.8% to 13.63% higher, as manufacturers and retailers absorbed the margins⁶⁷.

Although sales of both sweetened and diet beverages increased compared to the tax period, they remained below pre-tax baselines by 13% for high-sugar beverages and 24% for diet beverages⁶⁷.

Regarding whether the product reformulations (the replacement of sugars with alternative sweeteners) reversed or persisted on shelves after the March 2023 tax repeal, the exact market-wide status is **not established** in the provided research material. While overall sales volumes partially recovered, there is no documented industry or retail audit confirming whether manufacturers systematically reverted their product formulas to higher sugar content or maintained their alternative sweetener profiles.

E. Label Derivability

To translate clinical and regulatory findings into actionable, product-specific metrics, a clear distinction must be made between information that can be read directly off an Israeli product label and information that cannot.

Label-Derivable Parameters (Yes)

- **Sweetener Identity and Class:** Consumers can identify the specific sweeteners used in a product, as Israeli food regulations require all sweeteners to be declared in the ingredients list by their specific name (e.g., "סוכרלוז") or E-number¹³.
- **Aspartame Warnings:** The presence of aspartame is easily derivable, as the MoH mandates a highly visible warning label: "contains aspartame (a source of phenylalanine)" (מכיל אספרטיים (מקור של פנילאלין)) in a font size at least one-third the size of the product name¹³.
- **Polyol Laxative Warnings:** For products sweetened with sugar alcohols (such as sorbitol, maltitol, xylitol, and erythritol), standard labeling mandates the warning: "excessive consumption may cause increased intestinal activity" (צריכה מופרזת עלולה לגרום) לפעילות מעיים מוגברת)¹³.
- **D-Allulose Concentrations and Caloric Integration:** Under the 2024 MoH Novel Food

directive, D-allulose must be listed as "אלולוז" in the ingredients¹. Its caloric contribution must be calculated at 0.4 kcal/g as part of total carbohydrates, and its exact concentration in grams per 100 g of product must be declared directly adjacent to the nutritional facts table¹.

Non-Label-Derivable Parameters (No)

- Exact High-Intensity Sweetener Concentrations:** Except for D-allulose, the exact concentration of high-intensity sweeteners (such as sucralose, acesulfame-K, or steviol glycosides) is *not* label-derivable¹⁴. Israeli labeling laws do not require manufacturers to specify the quantitative milligram values per serving¹⁴. Consequently, a consumer cannot calculate their exact intake relative to the ADI directly from the label¹⁴.
- Microbiome and Glycemic Outcomes:** The potential for a sweetener to impair glucose tolerance or alter gut microbiota cannot be derived from a product label, as these physiological responses depend on individual biological factors and gut responder profiles²³.
- Endogenous vs. Exogenous Polyol Levels:** The physiological difference between endogenous synthesis of erythritol/xylitol (via the pentose phosphate pathway) and exogenous dietary exposure is not label-derivable, as fasting plasma markers require clinical laboratory verification⁴⁴.

F. Evidence Summary Table

The following evidence table synthesizes the parameters evaluated for each of the eleven sweeteners.

Sweetener Name	Hebrew Name	E-Number	Regulatory Status (EFSA / JECFA / Israeli MoH)	ADI (mg/kg bw/day)	Human Interventional Evidence Summary	Evidence Strength	Label-Derivable	Primary Citations
Steviol Glycosides	גליקוזידים של סטביל	E960	EFSA: Approved ¹⁸ JECFA:	4 (as steviol equivalents) ²⁰	RCTs show neutral glyce mic	Moderate	Yes (Identified as E960 or by chemi	[cite: 15, 20, 21, 22, 23, 24]

			Approved²¹ MoH: Approved¹⁵		impact²²; causes personalized, responder-specific microbiome shifts²³.		cal name) ¹⁴	
Sucralose	סוכרלוז	E955	EFSA: Approved¹⁵ JECFA: Approved²⁵ MoH: Approved¹⁵	15 (JECFA)²⁵; 5 (EFSA review)¹⁸	RCTs show impaired glycemic response and personalized gut microbiome changes in some individuals²².	Moderate	Yes (Declared as E955 or chemical name) ¹⁵	[cite: 15, 16, 18, 22, 23, 24, 25, 32]
Aspartame	אספרטם	E951	EFSA: Approved²⁶ JECFA: Appro	40²⁶	RCTs demonstrate neutral glyce	Moderate	Yes (Mandatory warning labels must	[cite: 13, 15, 16, 22, 23, 24, 26, 27, 72]

			ved ²⁷ MoH: Appro ved ¹⁵		mic impac t ²² and mild perso nalize d micro bial shifts ² ³ .		be displa yed) ¹³	
Acesu lfame -K	אצסול פאם ק"7	E950	EFSA: Appro ved ¹⁵ JECF A: Appro ved ²⁵ MoH: Appro ved ¹⁵	15 (JECF A); 9 (EFSA) ¹⁸	Insuffi cient clinical RCT data isolati ng its metab olic and micro biome effect s in huma ns ³⁰ .	Weak	Yes (Decla red as E950 or chemi cal name) ¹⁵	[cite: 15, 16, 17, 18, 30, 31]
Sacch arin	סחרין	E954	EFSA: Appro ved ¹⁵ JECF A: Appro ved ³² MoH: Appro ved ¹⁵	5 ¹⁵	Suez et al. (2022) RCT shows glyce mic impair ment ²³ ; separa te RCT shows	Moder ate	Yes (Decla red as E954 or chemi cal name) ¹⁵	[cite: 15, 22, 23, 24]

					completely neutral metabolic effects ²² .			
Erythritol	אריתריטול	E968	EFSA: Approved ³⁸ JECFA: Approved MoH: Approved ¹⁹	Not Specified ¹⁸	Highly glyceric-neutral ³ ; ex vivo models show prebiotic potential ³⁴ . Observational MACE signals exist but are subject to confounding ³⁶ .	Moderate (Glycemia) / Weak (Causal MACE risk)	Yes (Requires standard polyol laxative warning) ¹³	[cite: 13, 18, 19, 33, 34, 36, 37, 38, 42, 43]
Xylitol	קסיליטול	E967	EFSA: Approved ¹⁸ JECFA:	Not Specified ¹⁸	Highly glyceric-neutral ³ ;	Moderate (Glycemia) / Weak (Causal	Yes (Requires standard polyol	[cite: 13, 18, 19, 33, 39, 40, 42, 45,

			A: Approved MoH: Approved ¹⁹		reduces dental caries ³⁹ . Observational MACE signal is subject to endogenous confounding ⁴² .	al MACE risk)	laxative warning) ¹³	47, 48]
Sorbitol	סורביטול	E420	EFSA: Approved ¹⁸ JECFA: Approved MoH: Approved ¹⁹	Not Specified ¹⁸	Glycemic neutral ¹⁹ ; can cause osmotic diarrhea at high levels of exposure ¹³ .	Moderate	Yes (Requires standard polyol laxative warning) ¹³	[cite: 13, 18, 19]
Maltitol	מלטיטול	E965	EFSA: Approved ¹⁸ JECFA: Approved	Not Specified ¹⁸	Partially digested (2.0-2.4) ¹⁹ ; trigger	Moderate	Yes (Requires standard polyol laxative	[cite: 13, 18, 19]

			ved MoH: Appro ved ¹⁹		s mild glyce mic respo nse and potent ial laxatio n ¹³ .		warnin g) ¹³	
Monk Fruit	תמצי ת פרי הנזיר	N/A	EFSA: Not broadl y appro ved ¹⁸ JECF A: Under review ⁵² MoH: Autho rised as food ingred ient ⁶	Not Specifi ed ¹²	Small, short- term RCTs sugge st a 10-18 % reduct ion in postpr andial glyce mic excurs ions comp ared to sucros e ⁵⁰ .	Weak to Moder ate	Yes (Decla red as "מיץ מרוכז מפרי המונק " or similar) ⁶	[cite: 6, 7, 18, 49, 50, 51, 52, 53, 56]
Allulo se	אלולוז	N/A	EFSA: Under review ⁵⁷ JECF A: Proce ssing	Not Specifi ed ⁶⁰	Highly glyce mic-n eutral ³ ⁴ . ~70-8 0% absor bed in	Moder ate	Yes (Decla red as "אלולוז" with conce ntratio n declar	[cite: 1, 2, 34, 58, 60, 61, 62, 65]

			enzyme evaluated ⁶⁰		the small intestine and excreted in urine ⁵⁸ . High doses trigger gastrointestinal distress ⁶⁰ .		ed next to nutrition facts table) ¹	
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G. Contradiction Log

To support scientific objectivity, this log documents areas where the current scientific literature directly contradicts itself.

Saccharin and Glycemic Impact

- **Finding A:** Short-term saccharin consumption impairs glucose tolerance and alters gut microbiota composition in healthy adults²².
- **Finding B:** High-dose saccharin intake does not alter oral glucose tolerance, insulin sensitivity, gut hormone secretion, or microbial diversity²².
- **Reconciliation:** The metabolic and microbial effects of saccharin appear highly heterogeneous and sensitive to individual baseline microbiota configurations²².

Erythritol and Xylitol Cardiovascular Risk

- **Finding A:** High fasting plasma levels of erythritol and xylitol are associated with a significant increase in three-year incident MACE risk, and acute ^{30 g} ingestion enhances platelet reactivity in vitro and in vivo³⁶.
- **Finding B:** A five-week clinical intake of erythritol or xylitol does not alter vascular endothelial function, abdominal fat distribution, or glucose tolerance in obese subjects, and genetic markers do not show a causal link⁴⁰.
- **Reconciliation:** Observational cohort findings are confounded by endogenous synthesis of erythritol and xylitol through the pentose phosphate pathway under conditions of hyperglycemia, metabolic syndrome, and cellular ischemia⁴⁰. Platelet aggregation is observed at high bolus interventional doses, whereas normal dietary exposure (e.g., gum containing < 10 g/day consumed in divided doses) does not result in systemic

accumulation or thrombotic risk⁴⁵.

H. What We Cannot Say

To avoid alarmist narratives and ensure accuracy in consumer education, the empirical limits of the current scientific literature must be clearly defined.

- **No Causal Link to Real-World Cardiovascular Disease:** The scientific literature *cannot* assert that consuming erythritol or xylitol from foods or tabletop sweeteners causes heart attacks, strokes, or blood clots in humans⁴¹. The clinical associations identified by Witkowski et al. are observational and do not demonstrate causality⁴². They are heavily confounded by endogenous production through the pentose phosphate pathway, which is elevated in patients with underlying cardiometabolic risk⁴⁰.
- **No Universal Glycemic Harm:** The scientific literature *cannot* assert that alternative sweeteners universally cause glucose intolerance or diabetes²². The metabolic and microbiotic responses observed in interventional trials are highly individual-dependent and responder-specific, with steviol glycosides and aspartame showing glycemic neutrality in cohort studies²².
- **No Dietary Carcinogenicity of Aspartame:** The scientific literature *cannot* assert that standard consumption of aspartame causes cancer in humans²⁶. The Group 2B classification denotes a theoretical hazard based on limited, non-conclusive evidence, and JECFA has reaffirmed that typical dietary intake remains safe within the approved ADI of $40 \text{ mg/kg} \cdot \text{bw/day}$ ²⁶.
- **No Documented Post-Tax Reformulation Reversal:** The scientific literature *cannot* assert that the Israeli food and beverage industry systematically reverted its product formulas back to high-sugar recipes after the March 2023 sweetened-beverage tax repeal⁶⁷. This trajectory remains undocumented and is classified as not established⁶⁷.
- **No Empirical Proof of Israeli Consumer Confusion:** The scientific literature *cannot* assert that Israeli consumers confuse stevia, sucralose, and erythritol¹². There is no empirical data or published market research within Israel to support this claim.
- **No Pediatric Safety Clearance for Allulose or Monk Fruit:** The scientific literature *cannot* assert that D-allulose or monk fruit extracts are clinically safe or approved for consumption by infants, toddlers, or very young children¹. Israeli regulations explicitly prohibit their use in infant formulas, baby foods, and specialized clinical pediatric products¹.

I. Verification and Belief Statement

To maintain transparency, this section distinguishes between information verified through direct research material and beliefs held based on contextual indicators.

Verified Findings

The following regulatory and scientific facts have been verified:

- The Israeli Ministry of Health formally approved D-allulose as a Novel Food in February

2024, setting category-specific concentration limits¹.

- The Israeli Ministry of Health permitted the use of "concentrated monk fruit juice" as a food ingredient starting in 2017, with a precautionary exclusion in baby foods⁶.
- Short-term interventional RCTs show that alternative sweeteners alter the oral and fecal microbiome in an individual-specific manner, and that sucralose and saccharin can impair glucose tolerance in a subset of responders²².
- Fasting plasma erythritol and xylitol levels have been observationally associated with three-year incident MACE risk, but these markers are also endogenously produced by human cells via the pentose phosphate pathway under metabolic stress³⁶.
- The Israeli sweetened-beverage tax was implemented on January 1, 2022, and officially repealed on March 30, 2023⁶⁷.

Believed Findings

The following assertions are believed to be true based on contextual industry indicators, but are not verified by direct, empirical proof in the research material:

- It is believed that many manufacturers of sweetened beverages in the Israeli retail market maintained their reduced-sugar and sweetener-containing formulations post-repeal to avoid the red "high sugar" front-of-pack warning labels introduced in 2020⁶⁷. However, this cannot be verified as a market-wide trend due to a lack of comprehensive, post-market retail formulation audits.
- It is believed that some consumer confusion exists regarding the natural vs. synthetic origin of tabletop sweeteners in Israeli supermarkets¹². However, this cannot be confirmed due to a total lack of peer-reviewed consumer surveys or documented search query analyses within the State of Israel.

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